Directional differentiation and interpolation for Colombo's determinant problem

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Abstract

For distinct real numbers $x_1, \dots, x_N$, we give a self-contained proof that the matrix $[(x_j - x_i)^D]$ is nonsingular exactly when $N \le D + 1$ and at least one of N, D is even. The proof is organized around an auxiliary polynomial whose values have the signs of a putative kernel vector. Its real roots select directional derivatives that transform the associated homogeneous power sum into a strictly positive sum of even powers. A projective Rolle inequality then bounds the original root count. Ordinary interpolation treats opposite parities. When both parameters are even, a skew-symmetric derivative pairing annihilates the highest interpolation coefficient and lowers the bound by two. This construction gives a common proof framework for all nonsingular cases and closely follows an accompanying single-file Lean formalization. We explain its relation to Ma's Pfaffian positivity method and to the algebraic root-count method of Li, Tie, Wang, and Liu, including the common parity obstruction in the odd-exponent case.

Keywords. Difference-power matrix; real projective roots; directional derivative; Lagrange interpolation; formal verification.

1 The criterion and the proof architecture

For integers $N \ge 2$, $D \ge 1$, and pairwise distinct real nodes $x = (x_1, \dots, x_N)$, put

$$A_{N,D}(x) = \left[(x_j - x_i)^D\right]_{i,j=1}^{N}. \tag{1.1}$$

We prove the following complete criterion.

Theorem 1.1. *For the matrix in* (1.1),

$$\det A_{N,D}(x) \neq 0 \quad \iff \quad N \le D + 1 \text{ and } (N \text{ is even or } D \text{ is even}). \tag{1.2}$$

The criterion is already stated in Ma's treatment of Colombo's problem [1]. Li, Tie, Wang, and Liu give an algebraic proof of the remaining odd-exponent branch for even matrix size [2]. Both papers combine that branch with known results for even exponents. The present paper concerns a different organization of the proof, particularly the construction that closes the even-size, even-exponent case. The numerical criterion itself is not presented as a new result.

Suppose that $A_{N,D}(x)c = 0$ for a nonzero vector c. The associated polynomial

$$f(t) = \sum_{j=1}^{N} c_j (t - x_j)^D \tag{1.3}$$

vanishes at every node. We will bound its real roots by constructing a polynomial g with

$$c_j g(x_j) > 0 \qquad \text{whenever } c_j \neq 0. \tag{1.4}$$

The basic estimate is that $\deg g \le m \le D$ and $D-m$ even imply at most m projective real roots, provided at least two distinct nodes occur in the support. The same estimate establishes that f is nonzero, so no separate Vandermonde independence argument is needed in the sufficiency proof.

There are two applications. Interpolating $g(x_j) = c_j$ gives $\deg g \le N-1$ and settles opposite parities. In the remaining case, both N and D are even. The initial root bound forces a factorization $f = Pq$, where $P(t) = \prod_j (t - x_j)$ and q has no real root. A derivative pairing then produces an aligned interpolation polynomial of degree at most $N-2$. This last reduction is the main construction emphasized here.

2 Projective root counts and directional Rolle

For a nonzero real polynomial p with $\deg p \le d$, let

$$R_d(p) = Z_{\mathbb{R}}(p) + d - \deg p, \tag{2.1}$$

where $Z_{\mathbb{R}}(p)$ counts real roots with multiplicity. Equivalently, $R_d(p)$ counts the zeros of the degree-d homogenization

$$p^{[d]}(u,v) = v^d p(u/v)$$

on $\mathbb{RP}^1$, with multiplicity. The displayed expression is understood as a homogeneous polynomial, including at $v=0$. The term $d - \deg p$ is exactly the multiplicity at $[1:0]$. A nonzero form of degree zero has root count zero. For a nonzero homogeneous binary form F, we write $R(F)$ for this count.

For $w = (a,b) \ne (0,0)$, define

$$\mathcal{D}_w F = a\,\partial_u F + b\,\partial_v F.$$

Lemma 2.1 (Projective Rolle inequality)**.** *Let F be a real homogeneous binary form of degree $d \ge 1$. If $\mathcal{D}_w F \ne 0$, then $F \ne 0$ and*

$$R(F) \le R(\mathcal{D}_w F) + 1. \tag{2.2}$$

Proof. Nonvanishing of the derivative implies $F \ne 0$. An invertible real linear change of coordinates preserves projective root multiplicities and can send the direction of differentiation to ∂_u. In those coordinates, write

$$F(u,v) = v^\ell G(u,v), \qquad G(1,0) \ne 0, \qquad g(t) = G(t,1).$$

Then $\deg g = d - \ell$. The derivative is nonzero only if $d - \ell \ge 1$. Its factor at infinity has the same multiplicity ℓ, since $\partial_u G(1,0) = (d-\ell)G(1,0) \ne 0$. Thus

$$R(F) = \ell + Z_{\mathbb{R}}(g), \qquad R(\partial_u F) = \ell + Z_{\mathbb{R}}(g').$$

The ordinary Rolle theorem, with multiplicities, gives $Z_{\mathbb{R}}(g') \ge Z_{\mathbb{R}}(g) - 1$. Indeed, a root of multiplicity r contributes at least $r-1$ to the derivative, and between any two successive distinct roots there is an additional derivative root. If g has no real root, the inequality is immediate. This proves (2.2). □

Lemma 2.2 (A positive terminal derivative)**.** *Suppose that k successive nonzero real directional derivatives transform a degree-d form F into a nonzero scalar multiple of a form that is strictly positive on $\mathbb{R}^2 \setminus \{0\}$. Then $F \ne 0$ and $R(F) \le k$.*

Proof. The terminal form has no projective real zero. Its nonvanishing implies nonvanishing of every preceding form. Apply Lemma 2.1 backwards through the k derivatives. The case $k = 0$ is included. □

The directional derivative needed below has the simple form

$$\mathcal{D}_{(a,1)}(u - xv)^d = d(a - x)(u - xv)^{d-1}. \tag{2.3}$$

It lowers the exponent while multiplying the coefficient by a linear function of the node. Choosing that linear function is the essential degree of freedom in the argument.

3 A root bound from an aligned polynomial

Theorem 3.1 (Aligned polynomial root bound). *Let $x_1, \dots, x_N$ be distinct real numbers, and let $c \in \mathbb{R}^N$ have at least two nonzero coordinates. Let $0 \le m \le d$ with $d - m$ even. Suppose that a real polynomial g satisfies $\deg g \le m$ and*

$$c_j g(x_j) > 0 \qquad (j \in \operatorname{supp} c).$$

Then the polynomial $f(t) = \sum_j c_j (t - x_j)^d$ is nonzero, and

$$R_d(f) \le m. \tag{3.1}$$

Proof. The sign condition implies $g \ne 0$. Remove all real roots of g, counting multiplicities, to write

$$g(t) = Q(t) \prod_{\nu=1}^{\ell} (t - a_\nu), \qquad Q(t) \ne 0 \quad (t \in \mathbb{R}), \qquad \ell \le \deg g \le m.$$

By continuity, Q has constant sign. Choose $\sigma \in \{1, -1\}$ with that sign and put $h_0(t) = \sigma \prod_\nu (t - a_\nu)$. Then

$$c_j h_0(x_j) = c_j g(x_j) \, \frac{\sigma}{Q(x_j)} > 0 \qquad (j \in \operatorname{supp} c).$$

If $d - \ell$ is even, set $h = h_0$ and $k = \ell$. Otherwise choose $b > \max_j x_j$ and set

$$h(t) = (b - t) h_0(t), \qquad k = \ell + 1.$$

In the latter case $\ell < m$, because $\ell = m$ would contradict the assumption that $d - m$ is even. In either case we have

$$h(t) = \kappa \prod_{\nu=1}^{k} (t - b_\nu), \quad k \le m, \quad d - k \text{ even}, \quad c_j h(x_j) > 0 \ (j \in \operatorname{supp} c), \tag{3.2}$$

where $\kappa \ne 0$ and all b_ν are real. Repetitions among the b_ν are allowed. No b_ν is an active node, by the strict sign condition.

Consider the homogeneous form

$$F(u, v) = \sum_j c_j (u - x_j v)^d.$$

Repeated use of (2.3) gives

$$\begin{aligned}
\mathcal{D}_{(b_k,1)} \cdots \mathcal{D}_{(b_1,1)} F &= \frac{d!}{(d-k)!} \sum_j c_j \prod_{\nu=1}^{k} (b_\nu - x_j)(u - x_j v)^{d-k} \\
&= \frac{d!}{(d-k)!} \frac{(-1)^k}{\kappa} \sum_j c_j h(x_j)(u - x_j v)^{d-k}.
\end{aligned} \tag{3.3}$$

The last sum is strictly positive at every nonzero (u, v). Its nonzero coefficients are positive and its exponent is even. If the exponent is positive, two linear forms $u - x_i v$ and $u - x_j v$ with distinct nodes cannot both vanish at a nonzero point. If the exponent is zero, the sum is a positive constant. Lemma 2.2 now gives $F \ne 0$ and $R(F) \le k \le m$. Dehomogenization identifies this statement with (3.1). ☐

Remark 3.2. The theorem uses the degree of a polynomial encoding the coefficient signs, not merely the number of summands. Its terminal positivity proves nonvanishing as well as a root bound. It is a formulation of a classical sign-control mechanism, closely related to the root-count perspective of Sylvester and Reznick [3]; no claim of a new general rule of signs is needed for its use here.

4 Proof of the complete determinant criterion

4.1 The two necessary conditions

Let V be the $N \times (D+1)$ matrix with $V_{ik} = x_i^k$, indexed by $0 \le k \le D$. Let C be the $(D+1) \times (D+1)$ matrix whose only nonzero entries are

$$C_{k,D-k} = (-1)^k \binom{D}{k}.$$

The binomial theorem gives

$$A_{N,D}(x) = VCV^{\mathsf{T}}, \qquad \operatorname{rank} A_{N,D}(x) \le D+1. \tag{4.1}$$

Thus $N > D+1$ forces singularity. Also

$$A_{N,D}(x)^{\mathsf{T}} = (-1)^D A_{N,D}(x).$$

If N and D are both odd, taking determinants gives $\det A = -\det A$, hence $\det A = 0$. These are precisely the two obstructions on the right side of (1.2).

4.2 The kernel polynomial and its support

For the rest of the proof assume $N \le D+1$, and suppose, for a contradiction, that $Ac = 0$ for some $c \ne 0$. The support of c contains at least two indices. Otherwise, if only c_j is nonzero, any row $i \ne j$ would give

$$(Ac)_i = c_j (x_j - x_i)^D \ne 0,$$

contradicting the kernel equation. For the polynomial (1.3),

$$f(x_i) = (-1)^D (Ac)_i = 0 \qquad (1 \le i \le N). \tag{4.2}$$

Consequently, once $f \ne 0$ has been established, $R_D(f) \ge N$.

4.3 Opposite parities

Suppose that N and D have opposite parity. Let g be the Lagrange interpolation polynomial satisfying

$$\deg g \le N-1, \qquad g(x_i) = c_i.$$

At every active node, $c_i g(x_i) = c_i^2 > 0$. Apply Theorem 3.1 with $m = N-1$. Its hypotheses hold because $N-1 \le D$ and $D-(N-1)$ is even. It follows that $f \ne 0$ and

$$N \le R_D(f) \le N-1,$$

a contradiction. This argument includes the threshold $D = N-1$ and both opposite-parity choices.

4.4 Even size and even exponent: interpolation loses a degree

Suppose now that N and D are even. Since $N \le D+1$, parity gives $N \le D$. Start with the same interpolation polynomial $g_0(x_i) = c_i$, but apply Theorem 3.1 with $m = N$. We obtain $f \ne 0$ and

$$N \le Z_{\mathbb{R}}(f) \le R_D(f) \le N.$$

All inequalities are equalities. In particular, $\deg f = D$, the nodes are all the real roots of f, and each is simple. We can therefore write

$$f = Pq, \qquad P(t) = \prod_{i=1}^{N}(t - x_i), \qquad q(t) \ne 0 \quad (t \in \mathbb{R}). \tag{4.3}$$

Set

$$w_i = P'(x_i) = \prod_{j \ne i}(x_i - x_j) \ne 0. \tag{4.4}$$

Differentiating (4.3) at a node gives

$$f'(x_i) = w_i q(x_i). \tag{4.5}$$

The exponent $D-1$ is odd. Exchanging the indices in a finite double sum therefore proves the identity

$$\sum_{i=1}^{N} c_i f'(x_i) = D \sum_{i,j=1}^{N} c_i c_j (x_i - x_j)^{D-1} = 0. \tag{4.6}$$

This identity is valid for every vector c; the kernel equation is not needed for the cancellation itself.

Interpolate the new data

$$g(x_i) = c_i f'(x_i) w_i, \qquad \deg g \le N-1. \tag{4.7}$$

The Lagrange formula reads

$$g(t) = \sum_i g(x_i) \frac{P(t)}{(t - x_i) w_i},$$

where each quotient $P(t)/(t - x_i)$ is a monic polynomial of degree $N-1$. Taking the coefficient of t^{N-1} and using (4.6),

$$[t^{N-1}]g = \sum_i \frac{g(x_i)}{w_i} = \sum_i c_i f'(x_i) = 0. \tag{4.8}$$

Hence $\deg g \le N-2$, with the usual convention for the zero polynomial.

Since q has no real root, $q(0)q(x_i) > 0$ for every i. The scaled polynomial $h = q(0)g$ has degree at most $N-2$, and

$$\begin{aligned} c_i h(x_i) &= q(0) c_i^2 f'(x_i) w_i \\ &= q(0) q(x_i)(c_i w_i)^2 > 0 \qquad (i \in \operatorname{supp} c). \end{aligned} \tag{4.9}$$

Theorem 3.1 applies again, now with $m = N-2$, because $D-(N-2)$ is even. It gives

$$N \le R_D(f) \le N-2,$$

the final contradiction. No ordering of the nodes is used in this construction. The case $N = 2$ is included: the last auxiliary polynomial then has degree at most zero.

Completion of Theorem 1.1. Equation (4.1) and odd-dimensional skew-symmetry give necessity. Under the stated conditions, the opposite-parity and even-even arguments exclude every nonzero kernel vector. A square matrix over $\mathbb{R}$ with trivial kernel has nonzero determinant, proving sufficiency. □

5 Comparison with the two existing approaches

5.1 Ma: positivity of a Pfaffian integral

Ma's odd-exponent proof [1] constructs a strict sign for the Pfaffian. For ordered $N = 2r$ nodes, its output is

$$(-1)^{\binom{r}{2}} \operatorname{Pf}(A_{N,D}(x)) > 0.$$

The proof passes through paired determinants of truncated powers, B-spline and Marsden factorizations, total nonnegativity, Volterra propagation, and a Beta–de Bruijn integral representation. Positivity on an explicit region then gives a strictly positive integral.

The proof above excludes a kernel vector through a polynomial root bound. Its positive object is the terminal even-power sum in (3.3). This avoids constructing the paired determinants and integral representation. Ma's formalization also establishes intermediate positivity statements and the explicit Pfaffian sign [4]; those are additional outputs beyond the nonvanishing interface formalized here.

5.2 Li, Tie, Wang, and Liu: a closely related root-count contradiction

Li et al. [2] associate a real binary form to a kernel vector and apply the Sylvester–Reznick bound relating real projective factors to the length of a power-sum representation. For N even and $D \geq N + 1$ odd, their argument has the shape

$$N + 1 \leq R_D(f) \leq |\operatorname{supp} c| \leq N.$$

Our corresponding argument has the shape $N \leq R_D(f) \leq N - 1$. These inequalities express the same parity obstruction: for nonzero f of degree at most D,

$$R_D(f) \equiv D \pmod 2,$$

because nonreal roots occur in conjugate pairs. When D is odd and N is even, either side of the root-count comparison can incorporate this parity improvement. The change from N to $N - 1$ alone does not distinguish a new mathematical obstruction.

The relevant difference lies in the construction of the bound. Their Lean proof of the required finite-sum version [5] translates one node to zero, reflects the polynomial, and differentiates. One term becomes constant and disappears; induction reduces the number of summands to two. In Theorem 3.1, the chosen directions instead encode signs, and differentiation ends at a positive sum of even powers. Both developments use one-variable real polynomials with a degree parameter to account for roots at infinity.

5.3 The even-even construction and formal scope

The papers [1, 2] cite existing even-exponent results. Their source repositories have different scopes. Ma's checked repository formalizes the odd branch. The checked version of Li et al.'s repository also contains the even branch [5]. In its even-even proof, the first root bound likewise yields (4.3). It then establishes strict sign alternation of kernel vectors of a reduced odd-order skew matrix, using an explicit tangent-node configuration and a continuous deformation. A polar derivative identity yields a contradictory dot product.

Equations (4.6)–(4.9) provide the alternative used here: the skew identity cancels a leading interpolation coefficient, which directly improves the root bound. This is a concrete simplification of the auxiliary construction required for this criterion. It does not include the separate alternating-kernel theorem proved in [5].

The relationship with prior work is consequently twofold. The odd branch belongs to the same root-count family as [2, 3], while the aligned-polynomial formulation and the even-even interpolation construction give a compact route through all parameter cases. No priority claim for the determinant criterion or the underlying classical root-count principle is made.

6 The accompanying Lean proof

The complete criterion is formalized in the single file `DifferencePowerSingle.lean`, using Lean 4.32.0 and mathlib v4.32.0. It contains 958 lines, including comments and blank lines, and imports only `Mathlib`. All definitions and lemmas specific to this problem occur in that file. Its final declaration is

```
theorem determinant_ne_zero_iff {N D : ℕ}
  (hN : 2 ≤ N) (_hD : 1 ≤ D)
  (x : Fin N → ℝ) (hx : Function.Injective x) :
  (Matrix.det (fun i j : Fin N => (x j - x i)^D)) ≠ 0
  ↔ N ≤ D+1 ∧ (Even D ∨ Even N)
```

This is the full biconditional for arbitrary parameters and injective real node functions, rather than a statement restricted to examples.

Mathematical step	Principal Lean declaration
Rank and parity obstructions	`necessary`
Kernel equations give roots	`kernel_gives_roots`
Projective Rolle estimate	`projective_rolle`
Theorem 3.1	`root_bound_of_sign_polynomial`
Opposite parities	`kernel_zero_of_opposite_parity`
Equation (4.6)	`derivative_pairing_zero`
Even-even interpolation	`kernel_zero_of_even_even`
Theorem 1.1	`determinant_ne_zero_iff`

The code represents a degree-d form by a polynomial in $\mathbb{R}[X]$ and counts its finite roots together with the degree deficit at infinity. Translation and polynomial reflection implement the coordinate changes needed for directional differentiation. The proof of the even-even case uses exactly the interpolation data in (4.7) and the leading-coefficient cancellation in (4.8).

The file was copied to a fresh directory and compiled with a search path containing only the pre-existing mathlib and dependency caches. Compilation completed successfully. The transitive axiom audit of the final theorem reported

`[propext, Classical.choice, Quot.sound]`.

There is no unfinished proof placeholder or additional mathematical axiom in the development. The source, its verification log, and a SHA-256 manifest are supplied with this manuscript. The source is unchanged from the verification performed before the subsequent literature comparison.

The checked external source versions were `9f57b40c7c` for Ma [4] and `b98c9e55c9` for Li et al. [5]. Their repositories were inspected for the comparison above; neither complete external project was rebuilt locally for this manuscript. Our formal verification claim concerns the accompanying criterion proof, not the bibliographical or workflow statements in this paper.

Code and reproducibility. The complete single-file Lean proof, pinned project configuration, original verification record, and manuscript sources are available in the public repository

https://github.com/HappyAny/colombo-determinant-lean.

The repository also includes this PDF and instructions for checking the proof with Lean and mathlib 4.32.0.

References

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[4] Q. Ma, *Colombo odd-exponent theorem in Lean 4*, source snapshot 9f57b40c7c, accessed 6 September 2026. https://github.com/hkjtsgmc79-boop/colombo-odd-lean.

[5] *Colombo1928*, the Lean repository accompanying [2], source snapshot b98c9e55c9, accessed 6 September 2026. https://github.com/ttieli/Colombo1928.

AI assistance and development time

We developed the mathematical proof and its Lean formalization in an interactive session with **GPT-6 Astra**, with reasoning effort set to **max**. According to our session record, this original development phase took **approximately 50 minutes**. For this original solution-and-formalization phase, **Codex was the sole harness**, **fast mode was disabled**, and **no special orchestration skills were used**. Our prompts were brief: solve the supplied problem without online search, formalize the proof in Lean, and provide a complete single-file derivation. We did not supply a detailed proof strategy in those prompts; the ordinary Codex instructions and tools remained in use. We carried out the original problem-solving stage without web searches, using an existing local Lean/mathlib installation. We subsequently consulted the papers and repositories cited above for comparison and manuscript preparation. The reported 50-minute duration refers to the original proof-development phase and excludes that subsequent literature work and the writing of this paper.

The scope of Astra's training data and prior knowledge is unknown to us, including whether related mathematics or texts were present. Prohibiting online search during the session does not establish the absence of such prior exposure. The timing and workflow observations are therefore **for reference only**, as an uncontrolled case report. This qualification concerns provenance and efficiency claims; the mathematical criterion is separately supported by the Lean kernel check.

The case raises an open question: as model capabilities improve, does the *marginal benefit of additional task-specific orchestration* decrease? A stronger model may perform more planning and error correction within the model, reducing the need for some external coordination. This single session does not establish that trend. Codex still supplied tool execution, context management, and access to verification feedback, while Lean checked the resulting proof. A useful comparison would vary both model and harness on the same tasks under matched resource budgets, and measure verified completion and cost over repeated runs.

In particular, the mathematical result and its complete Lean proof were obtained under an explicit prohibition on online search during the original solution and formalization.